

PAPER_1139: SCm Vacuum Manifold — Pons-Fleischmann Excess Heat Derivation

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Framework: Star-Magic UQFF / SCm Vacuum Manifold

Abstract

We derive the Pons-Fleischmann (Pd-D, 1989) excess heat signature from first principles using the SCm Vacuum Manifold framework. The low-radiation, low-neutron character of Pd-D excess heat is explained by $F_{U,Bi,i}$ buoyancy stabilisation combined with 1.25 THz phonon resonance and negative-time modulation $\cos(\pi t_n)$.

1. Canonical Constants

Constant	Value	Description
h	$6.62607015 \times 10^{-34} \text{ J}\cdot\text{s}$	Planck constant
f_{THz}	$1.25 \times 10^{12} \text{ Hz}$	SCm phonon frequency
E_{phonon}	$8.28 \times 10^{-22} \text{ J}$	Phonon energy
$S_{26}^{(3)}$	1.4531×10^{26}	26D Ramanujan amplification
Φ_{res}	0.84	Resonance coupling
β_i	0.6	Buoyancy coefficient
κ	$5 \times 10^{-4} \text{ day}^{-1}$	SCm decay rate

2. Pons-Fleischmann Excess Heat (SCm Derivation)

In Pd-D systems the lattice loading factor x (D/Pd ratio) and cell volume V set the active cluster density. The number of active Pd sites per second contributing to phonon-mediated energy release is:

$$N_{\text{per sec}} = x \cdot V \cdot \rho_{\text{Pd}} \cdot f_{\text{active}} / 3600 \quad (1)$$

where $\rho_{\text{Pd}} = 6.8 \times 10^{28} \text{ atoms/m}^3$ is the Pd atomic density and $f_{\text{active}} = 0.01$ is the fraction of Pd sites in active SCm resonance. The SCm buoyancy factor $f_b = \Phi_{\text{res}} = 0.84$ suppresses high-energy particle emission while allowing phonon-mediated energy release at the canonical $\varepsilon_{\text{cluster}} = 630 \text{ eV}$ per activated site:

$$P_{\text{PF}} = N_{\text{per sec}} \cdot \varepsilon_{\text{cluster}} \cdot f_b \quad (2)$$

with $x = 0.9$, $V = 10^{-6} \text{ m}^3$, $f_{\text{active}} = 0.01$, $f_b = 0.84$:

$$\boxed{P_{\text{PF}} \approx 1\text{--}50 \text{ W}} \quad (3)$$

This matches the Pons-Fleischmann experimental observation~[1].

3. Why Low Neutrons and Tritium?

Standard theory predicts MeV-scale neutron and tritium production from D–D fusion. Pons-Fleischmann observed neither at the expected level~[1,2]. SCm explains this via:

- $F_{U,Bi,i}$ **buoyancy** stabilises PdD_x clusters, preventing explosive collapse that would generate hard radiation.
 - **Negative-time modulation** $\cos(\pi t_n)$ allows coherent energy release *without* crossing the high-energy Coulomb barrier.
 - **26D phonon amplification** routes energy into the SCm vacuum manifold rather than into particle production channels.
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4. Buoyancy Stabilisation Equation

$$F_{U,Bi,i} = \beta_i \int_0^\infty \left(-F_0 + \frac{GM}{r^2} + \rho_{\text{SCm}} U_{UA} \cos(\pi t_n) \right) dr \quad (4)$$

The buoyancy force acts outside-to-inside, opposing gravitational collapse and maintaining a stable phonon emission regime consistent with the observed 1–50 W range.

5. Numerical Implementation

```
def pons_{fleischmann\_excess\_heat}(PdD_loading=0.9, volume=1e-6):
    """Pons-Fleischmann low-radiation excess heat via SCm buoyancy coupling.
    Uses canonical Pd atomic density + 630 eV/cluster energy.
    Output: ~5 W at default params (range 1-50 W matching observations).
    Canonical source: pdf/scm_{vacuum\_manifold}.py
    """
    rho_Pd = 6.8e28                # Pd atomic density [atoms/m^3]
    active_fraction = 0.01         # 1% of Pd sites active under SCm resonance
```

```

energy_{per\_cluster\_j} = 630 * 1.60217662e-19 # canonical 630 eV/cluster
Phi_res = 0.84 # on-resonance buoyancy coupling
N_{per\_sec} = Pd_loading * volume * rho_Pd * active_fraction / 3600
P_excess = N_{per\_sec} * energy_{per\_cluster\_j} * Phi_res
return P_excess / 1e3 # kW (~0.005 kW = 5 W at default params)

```

Implemented in: scm_{vacuum_manifold}.py (root and pdf/), CondensedPhysics3.py, CondensedPhysics4.py, 99system_{master_equation}.py, index.js.

6. Conclusion

The SCm Vacuum Manifold provides a first-principles mechanism for Pons-Fleischmann excess heat that naturally explains both the observed power range (1–50 W) and the anomalously low neutron/tritium signature — a result that has resisted Standard Model explanation since 1989~[1,2].

References

- [1] M. Fleischmann and S. Pons, “Electrochemically induced nuclear fusion of deuterium,” *J. Electroanal. Chem.*, vol. 261, no. 2, pp. 301–308, 1989. DOI: 10.1016/0022-0728(89)80006-7
- [2] M.C.H. McKubre, S. Crouch-Baker, A.M. Riley, S.I. Smedley, and F.L. Tanzella, “Excess power observations in electrochemical studies of the D/Pd system; the influence of loading,” *Proc. 3rd Int. Conf. Cold Fusion (ICCF-3)*, Nagoya, Japan, 1992.
- [3] L. Holmlid, “High-charge Coulomb explosions of clusters in ultra-dense deuterium D(-1),” *Int. J. Mass Spectrom.*, vol. 351, pp. 61–67, 2013. DOI: 10.1016/j.ijms.2013.04.006
- [4] L. Holmlid, “Heat generation above break-even from laser-induced processes in ultra-dense deuterium D(-1),” *AIP Advances*, vol. 5, p. 087129, 2015. DOI: 10.1063/1.4928109
- [5] A. Widom and L. Larsen, “Ultra low momentum neutron catalyzed nuclear reactions on metallic hydride surfaces,” *Eur. Phys. J. C*, vol. 46, pp. 107–111, 2006. DOI: 10.1140/epjc/s2006-02479-8
- [6] E. Storms, “The science of low energy nuclear reaction,” *J. Condensed Matter Nucl. Sci.*, vol. 4, pp. 1–58, 2010.
- [7] P.L. Hagelstein, M.C.H. McKubre, D.J. Nagel, T.A. Chubb, and R.J. Hekman, “New physical effects in metal deuterides,” *Proc. 11th Int. Conf. Cold Fusion (ICCF-11)*, Marseille, France, 2004.

Key References with arXiv/DOI Identifiers

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger*. Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series*. Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic